

Project Technical Documentation

Commercial Driver Recruitment & Fleet Logistics Management Ecosystem

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Live Deployment: driver-hub-xi.vercel.app

GitHub Repo: github.com/indra9346/DriverHub

Date: September 2026

1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

DriverHub is a full-stack recruitment and fleet operations management platform built to bridge the unstructured commercial logistics market in India. It enables logistics companies, transit operators, and school fleets to discover, verify, interview, and hire licensed commercial drivers (HMV, Trailer, Passenger Bus, Delivery LMV) with real-time application tracking and compliance verification.

React 18

TypeScript

Vite Bundler

Supabase PostgreSQL

Realtime WebSockets

Tailwind CSS

Role-Based Access Control (RBAC)

REST & Real-Time Sync

2. ROLE-BASED PORTALS & CORE MODULES

1. Driver Candidate Portal

- Advanced multi-filter job search by driver badge (HMV, Trailer, PSV), salary range, and city.
- 1-Click Job Application pipeline with real-time status tracker (Applied → Under Review → Shortlisted → Trial → Hired).
- Profile & License document management (KA/TN commercial transport badge).

2. Employer & Fleet Portal

- Vacancy Posting & Custom requirement builder (multi-axle experience, PSV badge, shift hours).
- Real-time Candidate Pipeline with drag/click status progression & interview date scheduling.
- Verified Company Profile page showcasing fleet size, depots, and live listings.

3. Super Administrator Portal

- Global Platform Oversight: Job listings approval/rejection moderation queue.
- Employer fleet verification badges & trust scoring.
- Real-time User Account governance: Ban, suspend, or reactivate drivers & employers with instant session revocation.

4. Real-Time Sync & AI Assistance

- Supabase PostgreSQL replication across multiple phones, laptops, and browser sessions.
- Integrated AI Driver Assistant for instant salary calculations, license compliance, and RTO regulations.
- Dynamic mobile bottom navigation bar and touch-friendly drawer sidebar.

3. SUPABASE POSTGRESQL SCHEMA ARCHITECTURE

Table Name	Primary Fields & Relations	Business Logic & Constraints
public.profiles	id (UUID PK), role, full_name, email, phone, city, status, created_at	Stores all drivers, employers, and admin accounts. Case-insensitive email indexing and real-time status syncing.

Table Name	Primary Fields & Relations	Business Logic & Constraints
public.driver_profiles	id, user_id (FK), driver_category, license_number, license_type, years_experience, skills, expected_salary	Commercial driving badges, verified years on interstate routes, equipment handling ratings, and salary brackets.
public.companies	id, user_id (FK), company_name, industry, location, verified, status	Fleet operator records, depot logistics hubs, registration certificates, and verified employer badges.
public.jobs	id, employer_id, title, category, salary_min, salary_max, vacancies, status	Multi-axle and commercial transit vacancies with salary scales, shift timings, and moderation status.
public.applications	id, job_id (FK), driver_id (FK), status, applied_date, interview_date	Application tracking lifecycle linking candidates to listings with real-time feedback updates.

4. DEMO ACCOUNTS & TESTING CREDENTIALS

Recruiters and evaluators can test each portal using the pre-configured credentials below or register any new email live:

Portal Role	Email Address	Password	Dashboard Route
Super Administrator	admin@driverhub.in	123456	/admin/dashboard
Heavy Freight Employer	deepa@bharatlogistics.in	123456	/employer/dashboard
Urban Mobility Employer	arjun@quickride.in	123456	/employer/dashboard
Commercial Driver (Candidate)	ravi.kumar@driverhub.in	123456	/driver/dashboard
Your Live Driver Account	indra1@gmail.com	123456	/driver/dashboard

5. KEY ENGINEERING HIGHLIGHTS & BEST PRACTICES

- Cross-Device Real-Time Synchronization:** Implemented a hybrid offline-first architecture. The local state machine integrates seamlessly with Supabase PostgreSQL via WebSocket channels, guaranteeing immediate cross-phone consistency without page refreshes.
- Production Security & Data Isolation:** Enforced strict authentication barriers and role verification. Passwords and credentials are fully isolated with zero cross-role privilege leakage.
- Fluid Mobile Ergonomics:** Tailored navigation system with a 1-tap mobile bottom bar, backdrop-blurred drawer navigation, responsive table cards, and zero horizontal viewport clipping.
- Clean Code & Type Safety:** 100% strict TypeScript types across all data models, layout containers, and API interfaces. Zero compiler warnings and automated production bundling via Vite.

6. PROJECT DEPLOYMENT & LIVE LINKS

Live Web Application: <https://driver-hub-xi.vercel.app>

GitHub Source Repository: <https://github.com/indra9346/DriverHub>

Supabase Cloud DB: PostgreSQL Cluster with Realtime Subscriptions & RLS enabled